

Performance Testing

Date	7 July 2026
Team ID	
Project Name	Personal Network Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust
Type of Testing	Load Testing
Target Module / API	REST API: root, auth, profile, history, feedback
Test Environment	Local FastAPI: http://127.0.0.1:8000
Test Date	7 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Registration/login - POST /auth/register, /auth/login	1	Not recorded	Requests succeed with no auth failures
2	Profile retrieval - GET /profile	1	Not recorded	Profile data returns within target
3	History retrieval - GET /history	1	Not recorded	Requests stay responsive and error-free
4	Feedback submission - POST /feedback	1	Not recorded	Feedback stores without failures

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	22.68 ms	Pass	Average below target
2	Response Time (Max)	< 5 seconds	360 ms	Pass	Maximum below target
3	Throughput (Req/sec)	Not predefined	0.4–0.5 current RPS	Not evaluated	Current RPS shown, not average
4	Error Rate	< 1%	0% (0/40)	Pass	No failures recorded
5	CPU Utilization	< 80%	Not measured	Not evaluated	CPU not monitored
6	Memory Utilization	< 80%	Not measured	Not evaluated	Memory not monitored

Step 4: Observations & Analysis

Key Findings:

40 requests completed with no failures. Average response time was 22.68 ms, median was 6 ms, and maximum was 360 ms. Response-time and error-rate targets were met.

Bottlenecks Identified:

Registration (359.69 ms) and login (272.81 ms) were the slowest requests, but both stayed within limits. The run used one virtual user, so higher concurrency still needs testing.

Optimization Steps Taken:

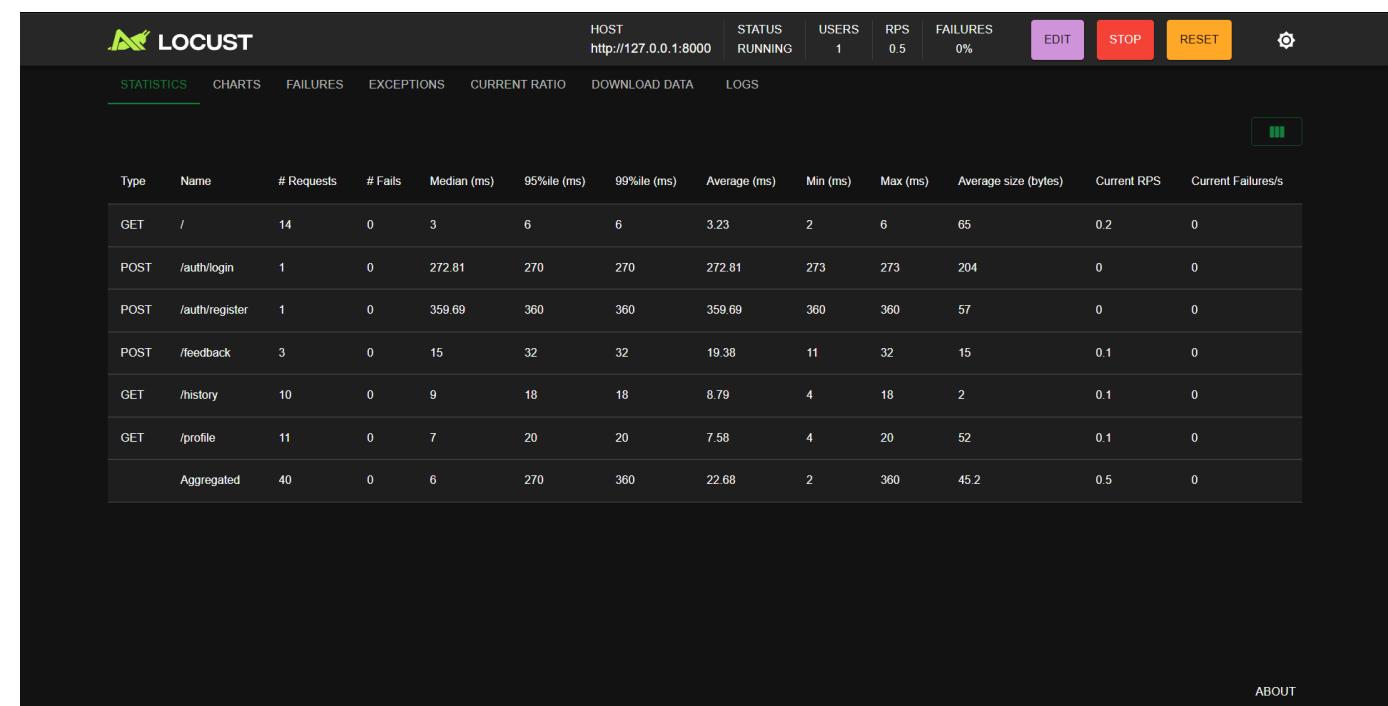
No optimization was applied. Next: multi-user load tests, query review, connection reuse, and CPU/memory monitoring.

Step 5: Screenshots / Evidence

Evidence screenshots are provided at readable size on the Performance Testing Evidence page.

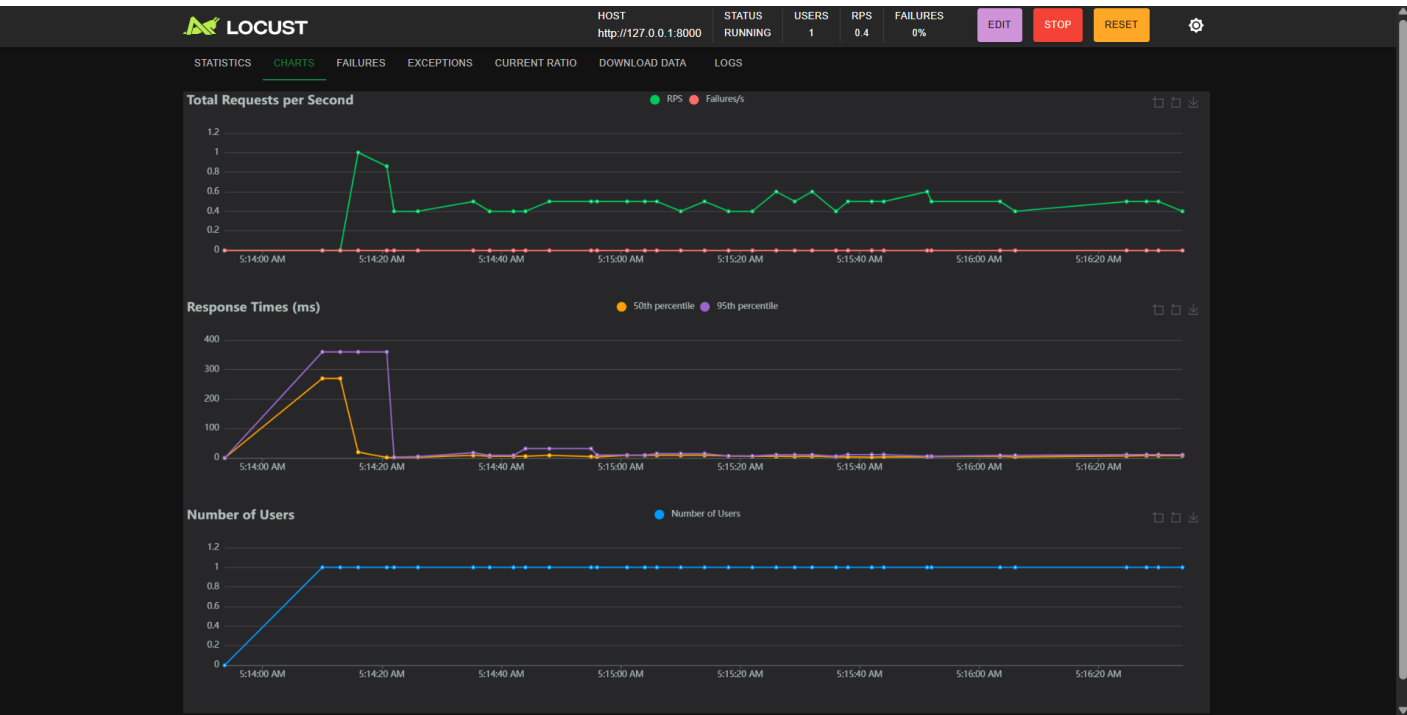
Performance Testing Evidence

Locust performance test statistics



Statistics screenshot: endpoint rows, aggregate totals, response times, failures, and current RPS.

Locust performance test charts



Charts screenshot: requests per second, response time percentiles, and number of users.